Confidently Wrong: Handing a Coding Agent the Database Anyway

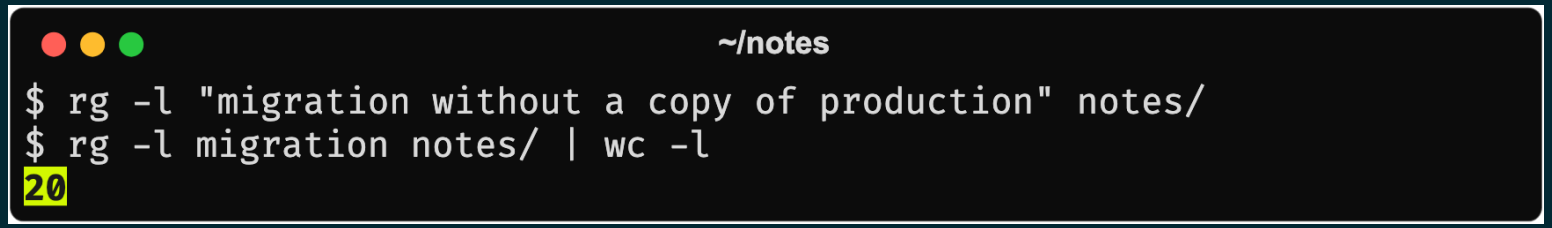
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All Things Open 2026, Databases track

A year of daily notes is 220 Markdown files

~/notes

```
$ ls notes/
2026-07-08.md  2026-07-23.md  2026-08-07.md  2026-08-24.md  2026-09-09.md  2026-09-28.md
2026-07-09.md  2026-07-24.md  2026-08-10.md  2026-08-25.md  2026-09-11.md  2026-09-30.md
2026-07-10.md  2026-07-27.md  2026-08-11.md  2026-08-26.md  2026-09-14.md  2026-10-01.md
2026-07-13.md  2026-07-28.md  2026-08-12.md  2026-08-27.md  2026-09-15.md  2026-10-02.md
2026-07-14.md  2026-07-29.md  2026-08-13.md  2026-08-31.md  2026-09-16.md  2026-10-05.md
2026-07-15.md  2026-07-30.md  2026-08-14.md  2026-09-01.md  2026-09-17.md  2026-10-06.md
2026-07-16.md  2026-07-31.md  2026-08-17.md  2026-09-02.md  2026-09-18.md  2026-10-07.md
2026-07-17.md  2026-08-03.md  2026-08-18.md  2026-09-03.md  2026-09-21.md  archive
2026-07-20.md  2026-08-04.md  2026-08-19.md  2026-09-04.md  2026-09-23.md
2026-07-21.md  2026-08-05.md  2026-08-20.md  2026-09-07.md  2026-09-24.md
2026-07-22.md  2026-08-06.md  2026-08-21.md  2026-09-08.md  2026-09-25.md
$ find notes -name "*.md" | wc -l
220
```

rg can't find a note I only half remember



I can ask my coding agent to look. It works, and it spends tokens I'd rather spend on code.

MariaDB

Why not have a coding agent build the fix?

Good ideas are always crazy until they're not.

Larry Page

I wrote one page about what I want

talk/notes-app-spec.md

```
## Why

I keep my notes as Markdown files in a folder, one file per day, named
with a date stamp. After a year there are hundreds of files. So, each
quarter, I summarize the work and move that quarter's files into an
archive folder.

Finding things is a problem. `rg` and `fzf` find exact words, but I
rarely remember the words. I remember that a note was good, that it has
the facts I need, and roughly what it was about. Asking a coding agent
to dig through the files works, but it burns tokens I would rather
spend on code.

## Stack

- MariaDB 12.3
- Everything runs on my laptop with no API keys. I don't want my notes
to leave my laptop.
```

The spec lists what to build now and what can wait

talk/notes-app-spec.md

```
## Must have

1. Import every note, current and archived
2. Pick up new and changed notes when I run the import again
3. Search by meaning, and show the best matches with their title, date,
and folder
4. Search for exact words too
5. Open a note and render its Markdown
6. Open the note in `nvim` if I hit enter
7. Search from the command line
8. Keybindings like `lazygit`, and the bottom bar is always visible

## Nice to have

1. `cron` automation
2. MCP server access
3. Standardized logging set through debug levels
```

The spec ends with a check the app must pass

talk/notes-app-spec.md

```
## Done when

- Every note file is in the database
- Searching for "testing a migration without a copy of production" shows my
April 14, 2026 note in the top three
```

The April 14 note never uses those words.
If the app finds it, the app searches by meaning.

First try: Claude Haiku 4.5 designs the database

Three prompts, then I run its SQL on a real MariaDB 12.3 server

PROMPT 1

The schema

Design the MariaDB schema and save it as working/notes_app.sql

PROMPT 2

The search

Write the SQL to search my notes by meaning and by exact words

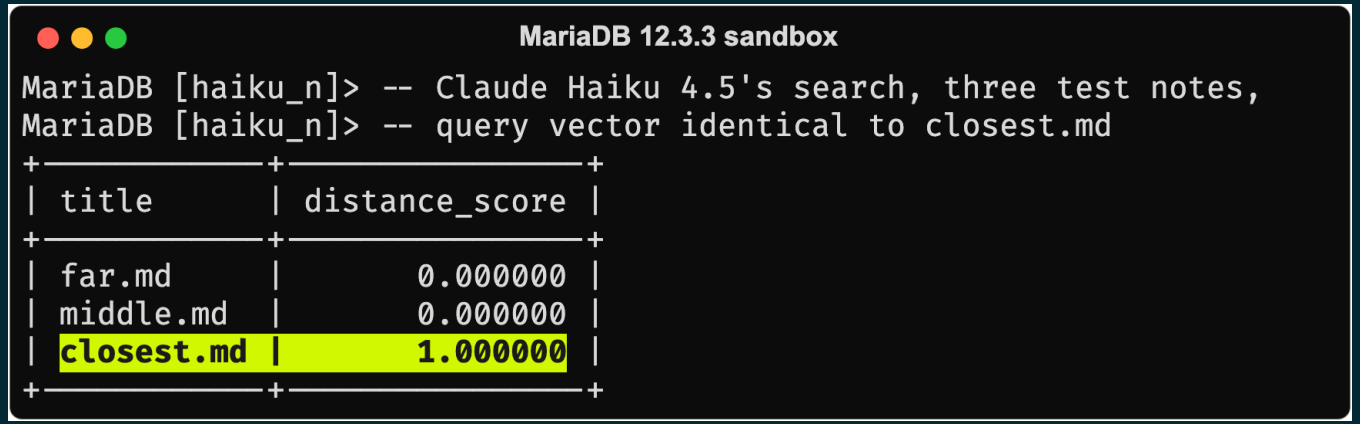
PROMPT 3

Reruns

Tell me how safe the schema is to run more than once

Its search ran but put the exact match last

MariaDB



The search used another database's distance operator.
In MariaDB, that operator asks whether two values are identical.

The agent called reruns safe, but they skip changes

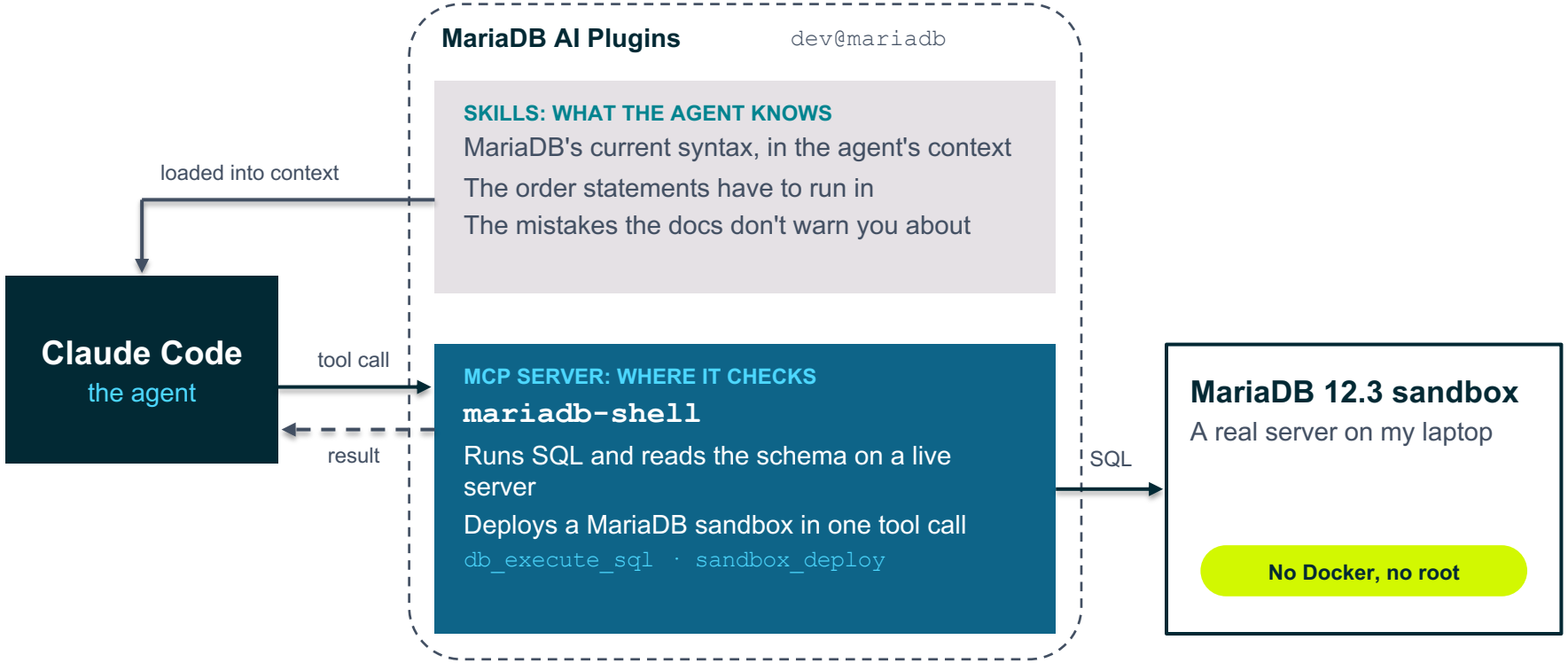
Claude Haiku 4.5, then MariaDB 12.3.3

● **Safety of running the DDL multiple times: ✅ Fully safe and idempotent.**

```
MariaDB [haiku_ns]> CREATE TABLE IF NOT EXISTS notes ( ... note_date DATE NULL ... );
MariaDB [haiku_ns]> SHOW WARNINGS;
+-------+------+--------------------------------+
| Level | Code | Message                        |
+-------+------+--------------------------------+
| Note  | 1050 | Table 'notes' already exists   |
+-------+------+--------------------------------+
MariaDB [haiku_ns]> SELECT COLUMN_NAME, IS_NULLABLE FROM information_schema.COLUMNS ...
+-------------+-------------+
| COLUMN_NAME | IS_NULLABLE |
+-------------+-------------+
| note_date   | NO          |
+-------------+-------------+
```

MariaDB's plugin adds knowledge and a live server

One plugin in Claude Code, with skills and an MCP server



MariaDB Public

Two commands to install, then config

```
/plugin marketplace add mariadb/ai-plugins
/plugin install dev@mariadb
```

```
$ mariadb-shell -- mcp setup --addPaths=~/notes-app
```

Folders the MCP server may read and write

```
$ mariadb-shell -- mcp setup --addConnection=demo@localhost:3306
```

Database servers it may connect to, with the password

The agent adds only the sandboxes it deploys.

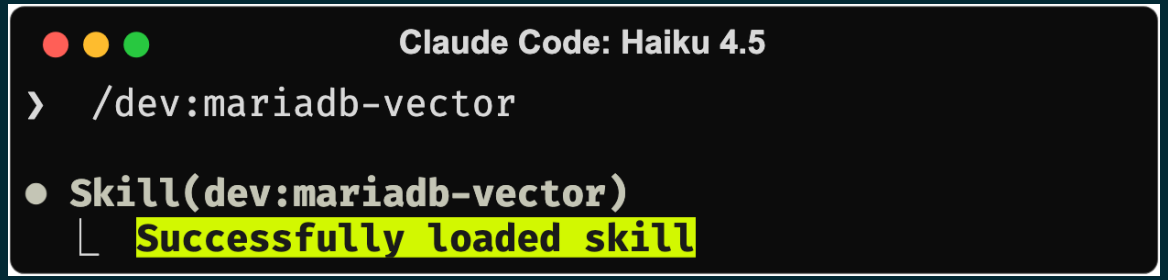
With the skills loaded, the exact match comes first

MariaDB 12.3.3 sandbox

```
MariaDB [haiku_n]> -- the search form from the skills-loaded run,
MariaDB [haiku_n]> -- same three notes, same query vector
+-------------+----------+
| title       | distance |
+-------------+----------+
| closest.md  |    0.000 |
| middle.md   |    0.293 |
| far.md      |    1.000 |
+-------------+----------+
```

Same agent, same prompt. Asked about reruns, it now says:
“Second run wipes the database.”

Check that the agent actually loaded the skill



Haiku 4.5 didn't load the skills on its own, even when my prompt asked it to. I loaded them myself.

Skills carry knowledge. Tools act on the server.

For the full build, I switched to Claude Opus 5.5

MariaDB

Claude Code: Opus 5.5

```
❯ Please run these steps
  Work in this repository and complete every step in order. Write your files to
  working/, and append a short record of each step to working/RUN_LOG.md.

  1. Design the MariaDB schema for the app that talk/notes-app-spec.md
     describes, and save the DDL as working/notes_app.sql.
  2. If no MariaDB 12.3 sandbox is running on port 3310, deploy one there with
     root password demo-pw and data directory working/sandbox. Run
     working/notes_app.sql on it via the MCP server.
  3. Report each table you created and what it holds.
```

Note: Name the spec file. Number the steps. Say “in order”.

Watch the transcript: four MariaDB Skill lines, before the agent wrote anything.

The agent built the app in about seven minutes

Two prompts, Claude Opus 5.5 with the MariaDB plugin

PROMPT 1 2:12

Designed the schema, deployed a sandbox, and ran the schema there

PROMPT 2 5:07

Built the terminal app and the search command, imported every note, and ran the “Done when” check

220
notes imported

351
sections embedded

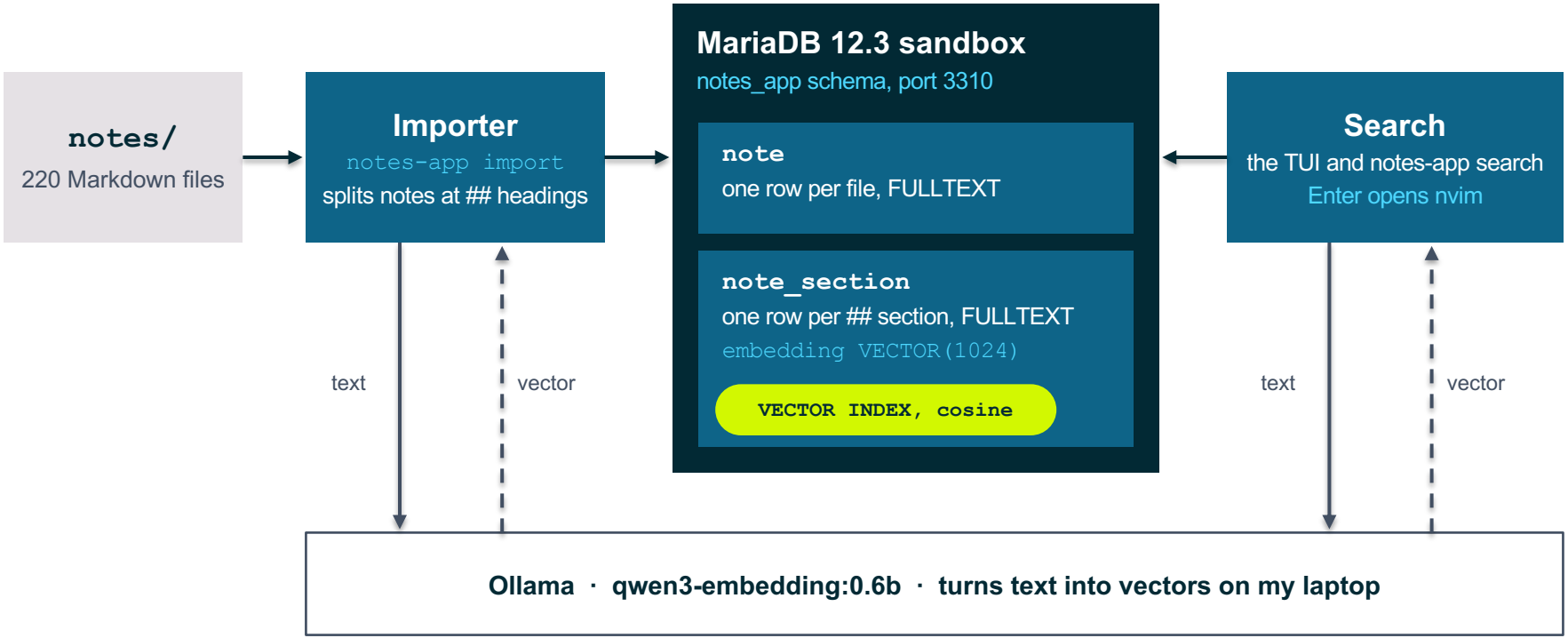
1,024
dimensions per vector

1
question for me, before it stopped an old server

MariaDB Public

The app searches my notes inside MariaDB

The notes, Ollama, the sandbox, and the app all run on my laptop



MariaDB Public

The agent tested its own work and fixed what it found

MariaDB

Claude Code: Opus 5.5

- **The editable install ended up empty since uv built the wheel with its own** backend instead of respecting hatchling's packages setting. I'll switch to uv_build and name the module explicitly to fix this.
  …
- **The TUI test passes fully, but it revealed a bug in exact-word mode: required** stopwords like "a" and "of" aren't stored in InnoDB's full-text index, so queries containing them return 0 results. I'll fix to_boolean_query to exclude those words.

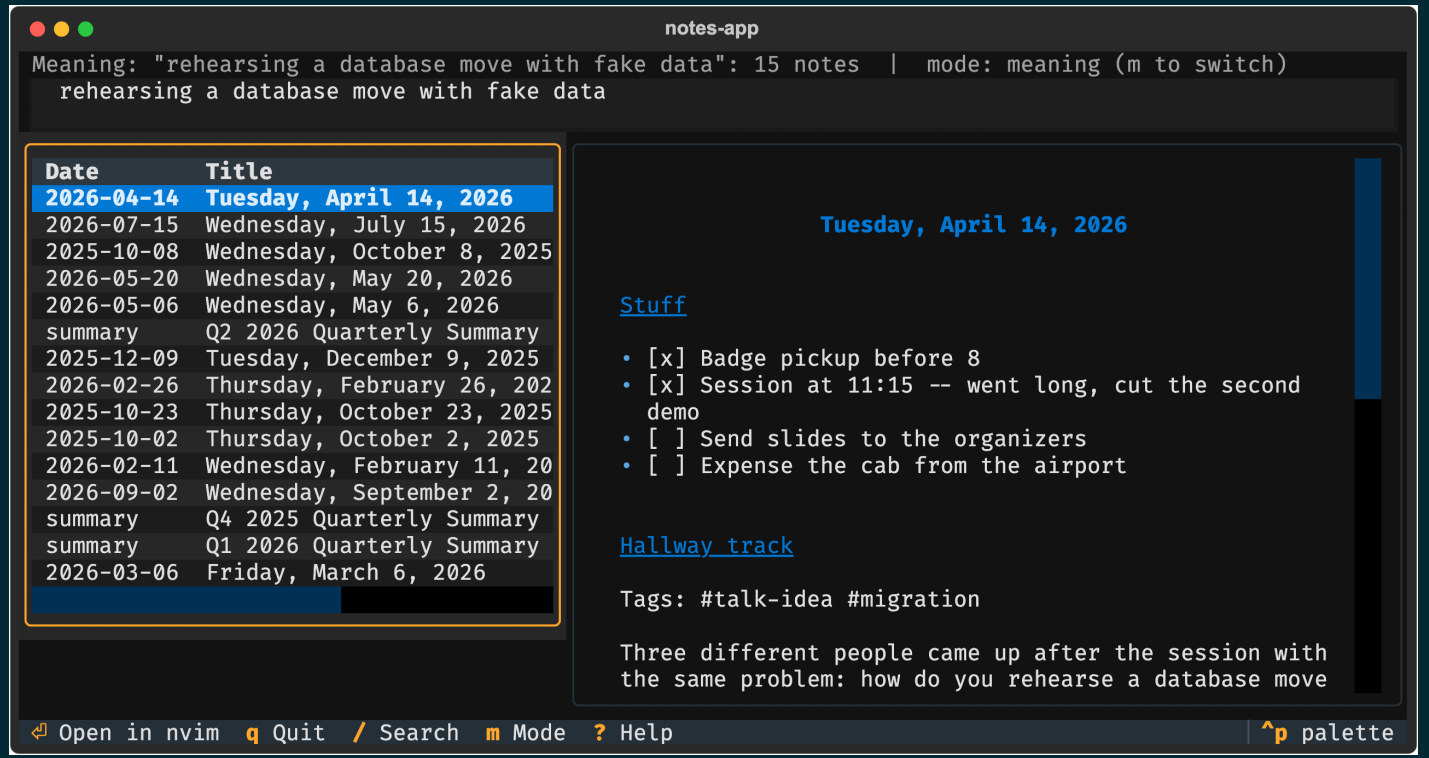
My check passes

~/notes-app

```
$ uv run notes-app search "testing a migration without a copy of production" -n 3
 1.  0.633  2026-04-14  Tuesday, April 14, 2026
              notes/archive/2026-Q2/2026-04-14.md  > Hallway track
 2.  0.628  summary     Q2 2026 Quarterly Summary
              notes/archive/2026-Q2/2Q2026.md  > APRIL 2026
 3.  0.623  2025-10-23  Thursday, October 23, 2025
              notes/archive/2025-Q4/2025-10-23.md  > Migrator demo notes
```

April 14 ranks 1st, just ahead of the Q2 summary that describes the same conversation.

Now I can search for what I remember



Add one line in CLAUDE.md to use my new tool

~/notes-app

```
$ cat CLAUDE.md
# Notes

To find one of my notes, ask the index first, with my words as the query:
`uv run notes-app search "<my words>"`. Read a note file only after the
search names it.
```

Without the line: grep and read, $0.43
With the line: ask the index first, $0.25

Same app, same question, one run each

HARD STOP: The harness stopped a delete on its own

```
DELETE FROM notes_app.account
```

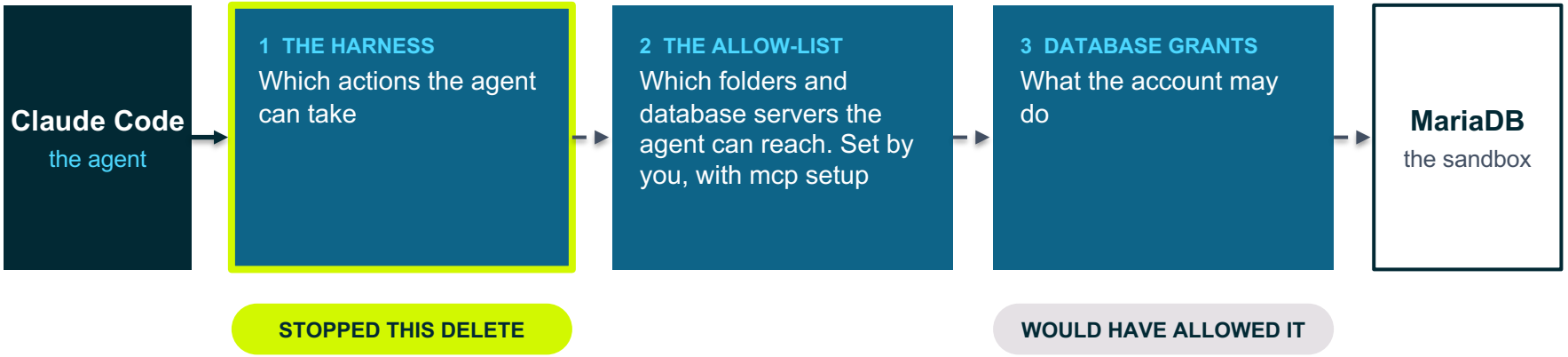
No WHERE clause. The foreign keys cascade to every other table.

“Permission for this action was denied by the Claude Code auto mode classifier. Reason: [Cloud Storage Mass Delete].”

Three layers decide what the agent can run

Each layer can say no on its own

```
DELETE FROM notes_app.account
```



What makes codeveloping with an agent easier

From one page of spec using a sandbox to checks that I run

SPEC

Adopt spec-driven development (SDD)

Write down what the app does, and the check it must pass

PROMPT

Name the files, number the steps

Tell the harness to do things “in order”

SKILLS

Load skills and plugins for your stack

Look for the Skill line to make sure the harness is paying attention!

SANDBOX

Give the agent a sandbox to try its work on

Always work in isolation when rapid-prototyping

Create snapshot clones for experimentation

CHECK

Run the check yourself

Provide feedback to the agent

MariaDB Public



Go build the tool you've been missing, on open source



github.com/maglietti/notes-app-demo
ai-plugins.mariadb.org